Identification of Boat Wakes Using Spectral Analysis and Neural Networks

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Introduction

- The objective of this project was to research a robust and adaptive method to track wake signatures for further quantitative analyses.
- Data was collected via a 16Hz RBR soloD pressure sensor deployed from September 22nd to October 11th of 2022.
- The sensor was deployed at the surface of the ICW just south of the CMS pier (Figure 1).

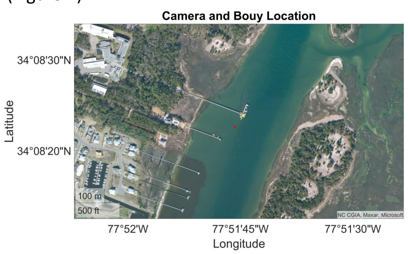


Figure 1 shows a satellite view of where the sensory buoy (red dot) was placed relative to the CMS Pier (northern pier) as well as the camera (yellow dot) used to train imgNet.

- Difficulties in tracking wake signatures comes from discerning wake signatures from noise.

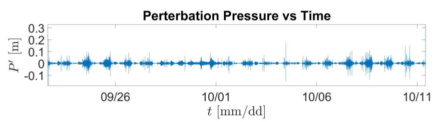


Figure 2a (top) shows the raw pressure data received from the sensor plotted against time. Figure 2b (bottom) shows the data after high pass filtering to remove low frequency (tides and atmospheric effects.).

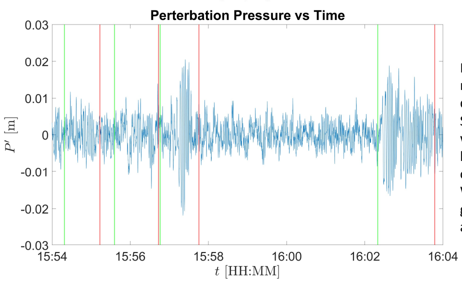


Figure 3 shows 10 minutes of pressure data from September 22nd with wake signatures that have been picked out by imgNet. Wakes begin at green lines and end at red.

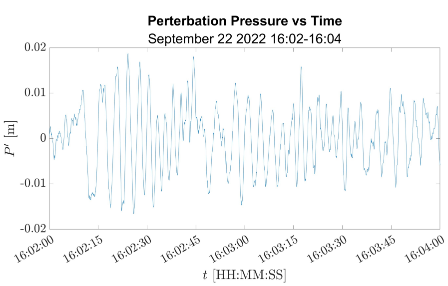


Figure 4 displays the last 32 second window shown in figure 3.

Method

- Perform a Short-Term Fast Fourier Transform(STFFT) in order to view waves in the frequency domain (Figure 4).
- An STFFT using a Hanning Window is better used here instead of Welches Method since it can isolate the wake signatures instead of average them out.

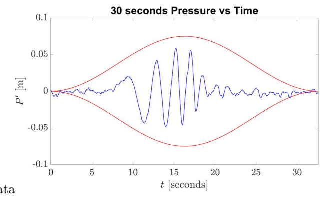


Figure 3(above) features a Hanning Window (32 seconds) in red and a wake signature in blue.

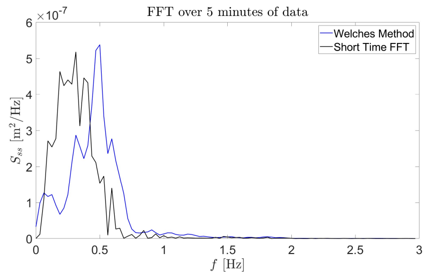


Figure 4 shows the magnitude of spectra graphed against cyclical frequency. Welches method (blue) fails to show the boat wakes at frequency .3 Hz as well as the STFFT(Red).

- By using window sliding and the Fourier transform we can create a spectrogram to view the wake signatures in both the frequency and time domain.

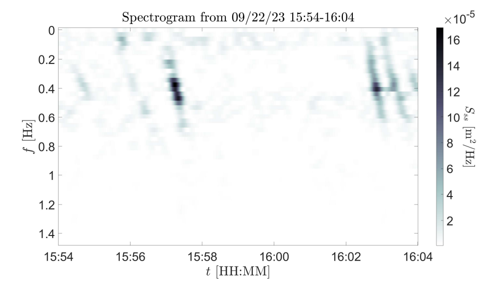


Figure 5 is a spectrogram that contains 5 wake signatures. 3 on the left and a double signature on the right.

- Now there's a method to accurately view wake signatures and pick out wake signatures from noise.
- The data will now be fed through a custom-trained neural network that will analyze the spectrogram as image data and pick out the waves.
- To label training images a video log was referenced in order to determine boats in a spectrogram.
- In order to prepare the data for analysis the spectrogram is converted into black and white. The color range is also limited which causes noise and wake signatures to be of similar shades.

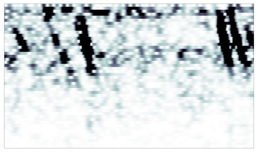


Figure 6 (left) shows what the neural network sees when it receives data.

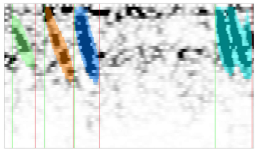


Figure 7 (right) shows the image after it has passed through the neural network.

Results

- By using this network, we can look at how wake energy and wind energy compare. Note that Hurricane Ian occurred on Sept 30th-Oct 1st, so the data is inaccurate.

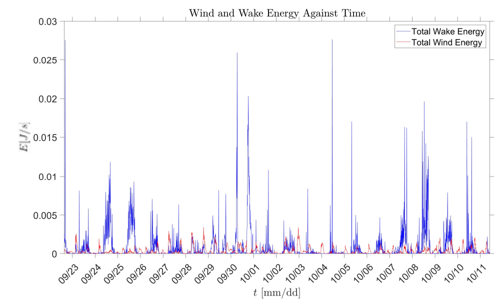


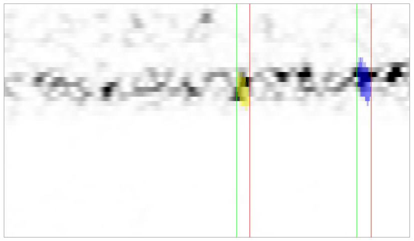
Figure 8 displays the total energies for wind(red) and wakes(blue) for every 10-minute spectrogram analyzed by imgNet.

- There are clear resurfacing peaks during the daytime as boat traffic drastically decreases at nighttime. There are also drastically more boats on weekends(9/24-9/25;10/8-10/9) then weekdays which can also be related to boat traffic.

Difficulties

- When moving forward retraining and restructuring of the neural network is required. There is still too much error in the networks selecting in order to create decisive and significant results. imgNet was only trained on 50 ten-minute images (roughly 300 wake signatures).

Figure 9 shows inaccurate classifications of wind noise as wake signatures during Hurricane Ian.



- Assuming a well-trained neural network, one could also calculate boat traffic demographics, characteristic features of wakes, characteristic features of wind waves and more.

References

Forlini, C., Qayyum, R., Malej, M., Lam, M. Y. -H., Shi, F., Angelini, C., & Sheremet, A. (2021). On the problem of modeling the boat wake climate: The Florida Intracoastal Waterway. *Journal of Geophysical Research: Oceans*, *126*(2). https://doi.org/10.1029/2020jc016676

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